

(5-year swap rate, 5 years forward). It is felt by some that the 5x10 rate strips away that component of a 10-year rate that is very much governed by Fed policy decisions and exclusively portends longer-term expectations of inflation, the growth of the economy, etcetera.

Former Fed Chairman Alan Greenspan, in his semiannual testimony before the Committee on Banking, Housing, and Urban Affairs, U.S. Senate, on February 16, 2005, alluded to this concept (i.e., that spot yields can be thought of as being constructed by a series of forward rates) to try to explain why the yield curve had flattened so much despite the Fed raising rates dramatically. Such flattening ran counter to previous tightening cycles.³⁹

In this environment, long-term interest rates have trended lower in recent months even as the Federal Reserve has raised the level of the target federal funds rate by 150 basis points. This development contrasts with most experience, which suggests that, other things being equal, increasing short-term interest rates are normally accompanied by a rise in longer-term yields. The simple mathematics of the yield curve governs the relationship between short- and long-term interest rates. Ten-year yields, for example, can be thought of as an average of ten consecutive one-year forward rates. A rise in the first-year forward rate, which correlates closely with the federal funds rate, would increase the yield on ten-year U.S. Treasury notes even if the more-distant forward rates remain unchanged. Historically, though, even these distant forward rates have tended to rise in association with monetary policy tightening.⁴⁰

Example

Figure 3.23 shows the 5x10 swap rate, the 10-year swap rate, and their difference in 2010. Although the 5x10 and 10-year swap rate are very highly correlated, their difference indicates that these two rates can vary significantly. As indicated by the data labels, the difference between the two rates widened out by 66 bps in a period of just over two months.

³⁹This phenomenon eventually led to conditions that became known as the *Greenspan Conundrum*. Despite a prolonged tightening cycle by the Fed, signs of economic growth and concerns over the risk of inflation, long-term rates did not rise, contrary to what history would suggest should have happened.

⁴⁰Testimony of Chairman Alan Greenspan at the Federal Reserve Board's semiannual Monetary Policy Report to Congress Before the Committee on Banking, Housing, and Urban Affairs, U.S. Senate, delivered on February 16, 2005. Transcript found on <http://www.federalreserve.gov/boarddocs/hh/2005/february/testimony.htm>

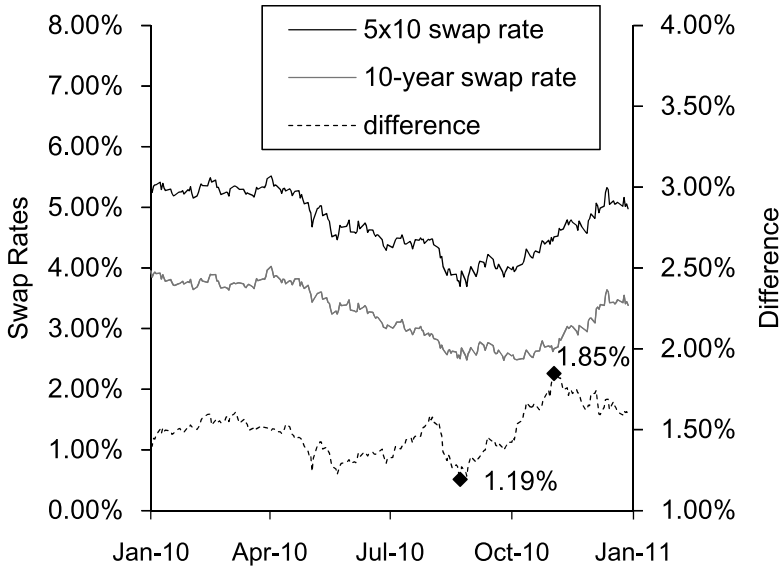


Figure 3.23: The 5x10 Swap Rate, the 10-Year Swap Rate, and Their Difference in 2010

As the graph indicates, the market had rallied significantly in the first part of the year, with the 10-year swap rate falling nearly 150 bps between January and August. The latter part of the year saw a large sell-off, with 10-year swap rates increasing approximately 100 bps from August to December. However, during this time, the 5x10 swap rate increased even more dramatically, moving in excess of 150 bps.

The 10-year swap rate can be bifurcated into the front five years and the back five years, which in our nomenclature is the 0x5 (the spot starting 5-year swap) and the 5x10 (the 5-year swap rate, five years forward). From August 26 to November 5, the 5x10 swap rate increased 82 bps, whereas the 10-year swap rate increased only 16 bps over this period. This gap can be explained by the movement in the 5-year swap rate. In fact, during this time the 5-year swap rate actually rallied nearly 24 bps.

Of course, the 5x10 swap rate is determined as a function of the 5-year and 10-year swap rates. But in the example we just considered, some market participants would interpret this as meaning that the sell-off in the long end was attributable more to changes in long-term prospects of inflation and the economy, rather than to perceived changes in Fed policy. □

Problem Set 3

For questions 1–7 below, consider the following swap curve:

Maturity	Swap Rate
1 Year	5.60%
2 Year	6.00%
3 Year	6.20%
4 Year	6.40%
5 Year	6.60%

All swap rates assume annual bond fixed payments versus 1-year LIBOR paid annual bond as well.

1. a) Bootstrap the curve and create a swap model. Use this model to compute the 2-year swap rate and the 5-year swap rate, verifying both swap rates as given in the above curve.

b) Referring to part a) above, suppose fictitiously that the two counterparties in each swap exchange principal at maturity. What is the value of the fictitious floating rate bond in the 2-year swap and in the 5-year swap?

c) Suppose now a counterparty is interested in receiving the fixed rate annual money. The floating side of the swap will continue to pay 1-year LIBOR annual 30/360. Use the swap model to compute the 2-year and 5-year swap rates paid annual money. For these purposes, assume that the fourth year of the 5-year swap has 366 days.

d) Provide a handcheck for your answers to part c), given the swap curve above.

2. a) Consider a 3-year swap, two years forward (i.e., a 2x5). Use your swap model to determine the par 2x5 swap rate. Use the handcheck method to verify that this is the correct 2x5 swap rate.

b) Referring to part a) above, suppose fictitiously that the two counterparties in the swap exchange principal at maturity. What is the value of the fictitious floating rate bond in the 2x5? This is a bond that pays three floating coupons (the first of which is received in three years and the last of which is received in five years) and principal at maturity. Explain your answer.

3. a) Suppose that two years ago, a client entered into a \$100mm 7-year swap with a dealer in which the client receives 5.80% versus 1-year LIBOR. What is the current mark-to-market value of the swap? Use the handcheck method to verify this answer.

b) This part is a continuation of part a) above. Suppose that the dealer is willing to unwind any swap by dealing 1 bp away from the relevant swap rate shown in the curve above, thereby incorporating bid/offer into the unwind value of the swap. What unwind fee is the dealer willing to show on the trade?

c) This question is unrelated to parts a) and b) above. Suppose that two years ago, another client entered into a \$100mm 4x7 year swap. That is, two years ago the client entered into a 3-year swap, 4 years forward. The client received a fixed coupon on this swap of 6%. What is the current mark-to-market value of the swap?

4. One of your clients has approached you to do an asset swap of a corporate bond that she just purchased. The bond mentioned pays a 7% coupon annually and matures in exactly five years (the coupon payment dates of the bond correspond with the payment dates in our stylized example of a swap). The bond was purchased at an annualized yield of 8.00%. The client is interested in swapping the bond into LIBOR plus a spread, thereby transforming her asset from a fixed rate bond into a floating rate bond.

The classic asset swap is a par transaction (and thus is sometimes referred to as a *par asset swap*). This means that if the purchase price of the bond differs from par, then there is an upfront payment recognized in the swap either from or to the client such that the sum of the purchase price of the bond plus the upfront equals par.

Use the bootstrap method to determine what spread to LIBOR you can provide your client.

5. An *amortizing swap* is one in which the notional of the swap declines over the life of the swap in accordance with some stated schedule. Coupon payments on each side of the swap are made on the same declining notional balances. Consider the following amortizing notional schedule:

Payment Date	Notional
1 Year	\$100mm
2 Year	\$80mm
3 Year	\$60mm
4 Year	\$40mm
5 Year	\$20mm

What is the par swap rate for this amortizing swap? That is, what fixed rate, when paid on the same declining notional balances as the floating side, gives the swap zero value today?

6. Suppose an investor wants to enter into two swaps at the same exact time. In particular, the investor would like to receive on \$100mm 2-year swaps and also receive on \$100mm 2x5 swaps.

a) What is the fixed rate in each of these two trades?

b) You presumably found two distinct fixed rates in part a). Suppose that immediately after the trades are completed, the investor indicates it would like to make changes to the terms of the trades. In particular, the investor would like to change the fixed rate in each swap and, furthermore, would like to constrain the fixed rate in each swap to be the same. Assuming that these changes are made in a way that does not affect the net present value of the two swaps as a package, what single fixed rate do you solve for?

c) What is the significance of the single fixed rate that you solved for in part b) in relation to the par swap curve for this question? Explain.

7. Suppose that earlier today a swap dealer paid fixed on \$100mm 4-year swaps to one of its clients at a rate of 6.50%. The client subsequently calls the dealer and reminds the dealer that three years ago, they had entered into a swap in which the client had paid fixed on \$100mm 7-year swaps at a rate of 6%.

The client is interested in doing a *pair off* of these two trades. That is, the client would like to know how much money (defined as the *pair off amount*) she needs to pay to or receive from the swap dealer today in order to simultaneously unwind both trades.

a) Given the current swap curve, what is this pair off amount and which counterparty pays it?

b) Suppose that the client does not act immediately on the level indicated in part a). Shortly thereafter the market sells off, and all swap rates are 5 bps higher. The client calls the dealer again and wants to deal at the same pair off amount you computed in part a). The dealer indicates that the pair off amount will change given the change in the market. The client insists that the pair off amount should be the same, since the net position has no risk. Should the dealer show the same pair off amount as you computed in part a)? Explain.

8. All of the swap rates in Table 3.10 are semi bond versus 3-month LIBOR quarterly money.

Use the relevant spot starting swap rates in this table to determine a hand-check for the par rate in a 5-year swap, two years forward (i.e., the 2x7). How close is this handcheck to the number in this table for the 2x7 rate?

9. In Appendix C, we look at a 2-year swap with an effective date of April 30, 2003, and a maturity of April 30, 2005. We indicate that on April 28, 2003, the par rate in this swap was 1.81643%. This assumes standard conventions of modified following with adjustment on both sides. If we change the conventions on the fixed side to following with adjustment, we find that the par fixed rate in the trade was 1.80671%, a difference of about 0.97 bps.

Suppose that the swap curve environment for this trade had been radically different. Suppose that at the time rates had been much higher. In particular, suppose that on April 28, 2003 the par swap rate for a 2-year swap effective April 30, 2003 and maturing April 30, 2005 was 10%, assuming modified following on both sides. Suppose now we consider changing the conventions on the fixed side of the swap to following with adjustment. Which of the following statements is true concerning the par swap rate with the fixed side following unadjusted?

- The rate differential will be 0.97 bps, the same amount as in the lower rate environment.
- The rate differential will be greater than 0.97 bps.
- The rate differential will be less than 0.97 bps.
- There is not enough info provided to determine the answer.

10. Consider the following two swap curves:

Swap Curve 1		Swap Curve 2	
Maturity	Swap Rate	Maturity	Swap Rate
1 Year	$x\%$	1 Year	$y\%$
2 Year	$z\%$	2 Year	$w\%$
3 Year	$a\%$	3 Year	$b\%$
4 Year	$p\%$	4 Year	$q\%$
5 Year	6%	5 Year	6%

Consider now a client who had previously received fixed on \$100mm of a 6-year swap. One year has now gone by, and as it stands today the swap is a 5-year swap.

True or False: The mark-to-market of the swap is the same under both swap curves regardless of the unspecified points on the curves since the 5-year rate in each curve is the same. Explain.

11. Consider the following two swap curves:

Swap Curve 1		Swap Curve 2	
Maturity	Swap Rate	Maturity	Swap Rate
1 Year	$x_1\%$	1 Year	$x_2\%$
2 Year	$y_1\%$	2 Year	$y_2\%$
3 Year	$z_1\%$	3 Year	$z_2\%$
4 Year	$w_1\%$	4 Year	$w_2\%$
5 Year	4%	5 Year	4%
6 Year	$a_1\%$	6 Year	$a_2\%$
7 Year	$b_1\%$	7 Year	$b_2\%$
8 Year	$c_1\%$	8 Year	$c_2\%$
9 Year	$d_1\%$	9 Year	$d_2\%$
10 Year	6%	10 Year	6%

Consider now the 5x10 swap rate.

True or False: Since the 5-year swap rate and the 10-year swap rate are the same in both curves, the 5x10 swap rate must be the same in both curve environments. Explain.

Chapter 4

Caps and Floors

There are two main types of plain vanilla options traded in the swaps market: caps and floors, on the one hand, and (single European) swaptions, on the other. In this chapter we will look at caps and floors. We will discuss how they are used by investors and how they are priced using a variant of the Black-Scholes model known as Black's formula.

4.1 An Introduction to Caps and Floors

A *cap* is a call on rates. Caps can be written on a variety of floating rate indexes, the most common of which is 3-month LIBOR. For example, the payoff on a cap on 3-month LIBOR depends on the formula

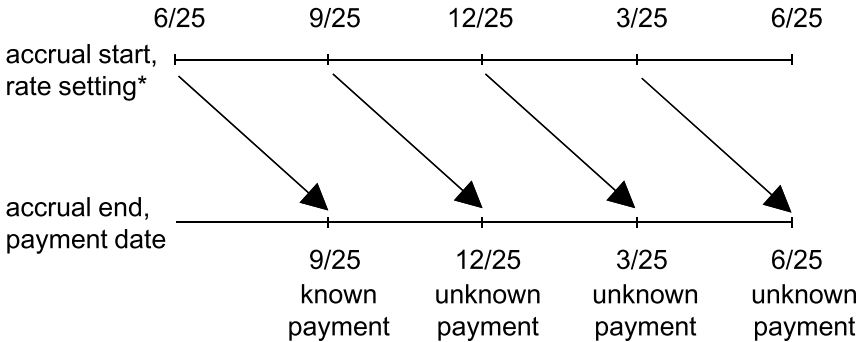
$$\max(3\text{-month LIBOR} - \text{Strike}, 0).$$

A *floor* is a put on rates. The payoff on a floor on 3-month LIBOR depends on the formula

$$\max(\text{Strike} - 3\text{-month LIBOR}, 0).$$

The holder of a cap receives a payment whenever LIBOR exceeds the strike, whereas the holder of a floor receives a payment whenever LIBOR is less than the strike.

The purchaser of a cap or floor on 3-month LIBOR is actually buying a series of options. LIBOR resets occur quarterly. A payment based on any in-the-money LIBOR setting occurs three months after the setting. That



* Rate setting is two London Business Days prior to start of accrual period.

Figure 4.1: Reset and Payment Conventions in Caps and Floors

is, the standard reset and payment conventions in a cap or floor concur with conventions on the floating side of a swap, as shown in Figure 1.5 and repeated here in Figure 4.1.

For example, consider an investor who purchases a 5-year spot starting cap on 3-month LIBOR on June 23. He is purchasing a sequence of 19 or 20 different options, depending on how we look at it. The *settlement date* of the cap is $T + 2$ New York and London Business Days after the trade date, which will be on June 25, assuming June 24 and 25 are both good business days. The settlement date is the day on which the premium for the cap will be paid by the buyer. It is also the day on which the first LIBOR reset will occur. Following the same convention as swaps, the LIBOR that will be used on any quarterly reset date will be the rate that is set two London Business Days prior. Thus, the first LIBOR reset that occurs on June 25 will actually use the rate setting two London Business Days prior to that date, which is June 23, and therefore is known on the trade date.¹ The first payment (which may very well be zero) based on this setting will occur three months thereafter on September 25. On September 25 we will again observe 3-month LIBOR (with a two-day lookback), and based on this setting, any in-the-money payment will be made three months thereafter.

¹Since the first reset in a spot starting cap is typically known at the time of trade, some people view the first reset and associated payment as not being an option, in which case the 5-year cap on 3-month LIBOR is viewed as a series of 19 options. Typically, the strike on a spot starting cap or floor is chosen such that the first reset is out-of-the-money.

We continue in a like manner each quarter, until the final payment in the 5-year cap is made five years after the settlement date.

Each individual quarterly option in the cap is referred to as a *caplet* (similarly, the individual options in a floor are known as *floorlets*). The payment made each quarter by a cap (i.e., the caplet payment) is equal to

$$\text{Notional} \times \text{Max}(3\text{-month LIBOR} - \text{Strike}, 0) \times (\text{Act}/360) \quad (4.1)$$

and by a floor (i.e., the floorlet payment) is

$$\text{Notional} \times \text{Max}(\text{Strike} - 3\text{-month LIBOR}, 0) \times (\text{Act}/360). \quad (4.2)$$

4.1.1 Cap-Floor Parity

The analog of put-call parity, cap-floor parity provides us with a means to relate cap prices to floor prices. It is easy to verify that²

$$\begin{aligned} & \text{Max}(\text{LIBOR} - \text{Strike}, 0) - \text{Max}(\text{Strike} - \text{LIBOR}, 0) \\ &= \text{LIBOR} - \text{Strike}. \end{aligned} \quad (4.3)$$

Let τ denote the day count fraction. Multiplying each side of Equation 4.3 by $\tau \times \text{Notional}$, we have

$$\begin{aligned} & \tau \times \text{Notional} \times \text{max}(\text{LIBOR} - \text{Strike}, 0) \\ & - \tau \times \text{Notional} \times \text{max}(\text{Strike} - \text{LIBOR}, 0) \\ &= \tau \times \text{Notional} \times (\text{LIBOR} - \text{Strike}). \end{aligned} \quad (4.4)$$

This expression says the payout from a caplet less the payout from a floorlet with the same strike is equal to the cash flow in an individual period resulting from paying fixed in a swap (known as a *swaplet*). Here it is understood that both the fixed and floating side of the swap have the same day count and payment frequencies as the caps and floors in question. The relationship in Equation 4.4 is true for each and every caplet, floorlet, and swaplet in the trade. Thus the payouts from a cap less the payouts from a floor are the same as payouts obtained from paying fixed in a swap. From this we conclude that cap-floor parity must hold:

$$\text{Value of Cap} - \text{Value of Floor} = \text{Value of Paying Fixed in Swap}. \quad (4.5)$$

²To show that this is true, suppose first that $\text{LIBOR} \geq \text{Strike}$. Then the left-hand side of Equation 4.3 is equal to $\text{LIBOR} - \text{Strike}$, which is of course equal to the right-hand side. Now suppose that $\text{LIBOR} < \text{Strike}$. Then the left-hand side is equal to $-(\text{Strike} - \text{LIBOR}) = \text{LIBOR} - \text{Strike}$, which is again equal to the right-hand side.